

University Degree in Telematics Engineering
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Bachelor Thesis

"Machine Learning-Based Predictive Modeling of Energy Prices"

Rodrigo De Lama Fernández

Emilio Parrado Hernández

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SUMMARY

Keywords:

DEDICATION

Dedicated to my late grandfather who was not able to see me become an engineer like himself

Here is to my amazing family

To my parents, that have always To my grandparents

To my girlfriend

To my friends and colleagues

To anyone and everyone that has supported me during my years in university

Here is to the next steps in life

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1. INTRODUCTION TO THE PROBLEM

- **1. Explanation of how the price of energy works in Spain**
 - a. Energy
 - b. Prices
- **2. Machine Learning - Lets run through some model overviews, which are available to choose, but without actually referencing anything**

Energy has increasingly become more important. Availability is key for a modern high functioning society. In Europe we are lucky to have a really advanced and reliable network. It is a key resource needed to succeed in the ecological transition that we are currently undergoing world wide. Electric mobility is a key element of this process. Electric cars, buses and trucks, need easily available and convenient sources of energy to recharge, to be able to play their role in society. An important factor in this ever growing necessity is the price of the available energy. But in recent times, that easy and affordable access has changed, and has become more unpredictable. Ever since the commencement of the war between Russia and Ukraine marked an inflection point in the mostly cyclical nature of electricity prices. Prices have gone up considerably since a couple years time, and it has been affecting everyone, from big consumers such as companies, to everyday people, having to pay a more expensive electricity bill at the end of the month. This has become a topic of great concern, with more and more people thinking about it often.

This led me to think about pricing, how it was structured, and what to expect as an end consumer. Could we plan our use of energy according to the price? Could it be predicted accurately? These were some of the questions I had that sparked my interest in looking into energy predictions.

In the initial discussions of this project, we talked about various ways to attack the problem at hand. We thought about a variety of systems but settled on a plan to explore the predictions with the more established methods of Machine Learning. This was done specially since it was a natural continuation of what was learned in the third year subject of Telematics Engineering, Modern Theory of Detection and Estimation.

The first idea we developed was to separate out the project in 24 blocks, one for each hour block to predict. The reason behind splitting the data in 24 blocks was simple, we assumed that data between the same time slot across neighbor days, would be similar, or would follow a certain trend.

Focusing on the innovation side of things, we decided to simplify the problem. Pivot-

ing from creating an algorithm to predict hourly prices, to splitting that into the 24 blocks, and predicting a single one. We decided on the 14th hour of each day, since irrespective of the day of the week, or holiday, it would always be a valley period.

This kind of granular breaking up of data is similar to what a Random Forest does in order to absorb more details and create higher resolution predictions. We made this assumption based on the idea that it would simplify the project substantially.

Talk about Emilio's background in investment banking at BBVA and his alorithms.

Talk about the idea of using technical analysis for feature engineering

2. BACKGROUND AND RELATED WORK

- **References**

- How is the price of electricity / energy set? What happens if new producers enter the market?
 - Articles that reference energy prices prediction
 - * As an inspiration for my own work:
 - 5-10 articles that do/talk about similar things
 - Energy consumption predictions
 - Energy prices in other European countries (Italy, Germany, UK, France)
 - Longer term predictions
 - * Predicciones de produccion
 - Describir tecnologias – explicar mas en detalle como funciona y enlaces a donde pueda averiguar el contenido
- Referenciar que metricas uso para medir mi error etc
 - Referencias de donde saco los datos
 - Referencias a que software utilizo referencias a documentacion de sklearn etc

How is the price of electricity set in Spain and Portugal?

What happens when new producers enter the market and sell to the energy pool?

Comment about more typical case studies focusing on energy consumption, since that is something more consistent and predictable

Errors: R^2

The science of technical analysis TA metrics used? SMA EMA ROC RSI

Talk about the intervals used:

$SMA_3, SMA_5, SMA_7, SMA_{14}, SMA_{30}, SMA_{60}, SMA_{90}, SMA_{180}, SMA_{360}$

$EMA_3, EMA_5, EMA_7, EMA_{14}, EMA_{30}$

$ROC_3, ROC_5, ROC_7, ROC_{12}, ROC_{14}, ROC_{30}$

RSI_5, RSI_7, RSI_{14}

3. PROPOSAL - THEORETICAL SYSTEM DESCRIPTION

Contar todos los pasos del pipeline de datos

Pipeline steps:

1. Data download from: <https://www.omie.es/es/file-access-list?parents=/Mercado%20Diario/1.%20Precios&dir=Precios%20horarios%20del%20mercado%20diario%20en%20Espa%C3%91a&readdir=marginalpdbc> [1]

Downloaded with a python script for the dates that were not available with an easy .zip file (for whole previous years)

2. The data has this shape:

```
MARGINALPDBC;
2018;01;01;1;28.1;6.74;
2018;01;01;2;33;4.74;
2018;01;01;3;32.9;3.66;
2018;01;01;4;28.1;2.3;
2018;01;01;5;27.6;2.3;
2018;01;01;6;24.6;2.06;
2018;01;01;7;20.1;2.06;
2018;01;01;8;19.9;2.06;
2018;01;01;9;19.84;2.3;
2018;01;01;10;19.9;2.3;
2018;01;01;11;19.9;2.3;
2018;01;01;12;19.9;2.3;
2018;01;01;13;23.6;2.3;
2018;01;01;14;25.1;2.3;
2018;01;01;15;23.6;5;
2018;01;01;16;24.6;5;
2018;01;01;17;25.1;5;
2018;01;01;18;27.6;8.85;
2018;01;01;19;27.6;15.93;
2018;01;01;20;28.1;22.02;
2018;01;01;21;32.9;20;
2018;01;01;22;28.1;21.95;
2018;01;01;23;28.1;23.52;
2018;01;01;24;27.6;16.35;
```

*

3. Data format: https://www.omie.es/sites/default/files/2024-06/Formato_ficheros_inf_pub_135_0.pdf

3. In page number 67, chapter 6.18 we may find the following information regarding the encoding format for the information: [2]

6.18 Precios marginales del mercado diario (MARGINALPDBC)

Fichero con los precios marginales del Mercado Diario para cada una de las horas.

Nombre del fichero: marginalpdbc_aaaammdd.v donde aaaammdd corresponde a la fecha de sesión y v es la versión del fichero.

Descripción de los campos:

CAMPO DESCRIPCIÓN VALORES VÁLIDOS

Año Año I4 - 20XX

Mes Mes I2 - 1 a 12

Día Día I2 - 1 a 31

Hora Hora I2 - 1 a 25

MarginalPT Precio marginal zona Portuguesa F8.2 -
-99999.99 a 99999.99

MarginalES Precio marginal zona Española F8.2 -
-99999.99 a 99999.99

4. Data organization

We will transform the data into a CSV format easier to use with the Pandas library to later create a DataFrame. We will maintain the same information, but in a encoding more suitable for Pandas. For that, the date and time must be transformed to a format such as Datetime which will transform the information from 2018;01;01;14; ... to 2018-01-01 14:00:00

5. Data retention of only the relevant info time and date, and MarginalES - drop MarginalPT

6. Data cleanup: Check for missing or duplicate entries. since Daytime Savings introduces duplicate hours some days of the year: INSERT some dates in the dataset that have the duplicated entries

7. Explain all blocks of the system? Data download: Data ingest

Contar los bloques empleados en cada parte – como junto las piezas para construir el sistema – porque uso las piezas que uso

Hablar de implementaciones como que librerías, que scripts etc – no es muy necesario, mas importante la idea

This chapter will outline the theoretical design of the system developed for the project. It will outline the complete pipeline constructed for the system: starting from the data preparation stage, continuing with the data preparation to the final results.

Data Preparation

Matrix Creation

Mini-models - sliding window approach

Technical indicators

ML blocks Linear Regression Linear combination of every feature Lasso Regression
Can turn unimportant features into 0s Random Forest Feature engineering

Pandas, NumPy, scikit-learn

4. EXPERIMENTATION

a. Data Description – datos del OMIE

b. Set Up Explanation – How do I use the previously explained system - what parameters did I modify to obtain the final results. Feature engineering, Alpha variation testing in Lasso, Tree depth and como lo usas en el sentido de tocar cosas, parametros como numero de arboles profundidad, el Alpha del lasso

c. Results a modo tabla unica comparando todas las iteraciones de cada tecnologia, y luego una conjunta con la mejor de todo (Si me hace falta mas, pues al apendice)

d. Low level discussion – conclusión que le contaria a un compañero de como he hecho todo esto

5. REGULATORY FRAMEWORK

Not essential for this investigative project because we don't go to market

Habria que hablar de las normas, pq si alguien lo usa es para ir a mercado

I need to talk about laws, what data is permissible to use and how. This would be essential for a project that does indeed go to market

For us its not that relavant

Software & Licenses: Visual Studio Code Python Scikit-learn Pandas NumPy

macOS / Windows / Ubuntu

6. SOCIO-ECONOMIC ENVIRONMENT

- a. Energy prediction
- b. Why is this relevant
- c. Shutdowns
- d. Budget - Price breakdown of the investigation
- e. Gantt diagram representing time spent - Tiempos de diseño e innovacion. Amortizar nuestras horas de trabajo mias y emilio, ordenador y poco mas

7. CONCLUSIONS

- a. Recap
- b. Revisit the objectives Predicting the prices in an accurate manner using Machine Learning algorithms and technical analysis
- c. Future work Less error

BIBLIOGRAPHY

- [1] OMIE, *Precios horarios del mercado diario en españa*, <https://www.omie.es/es/file-access-list?parents=/Mercado%20Diario/1.%20Precios&dir=Precios%20horarios%20del%20mercado%20diario%20en%20Espa%a1sa&readdir=marginalpdbc>, Accessed on May 27, 2025, Jun. 2024.
- [2] OMIE, *Formato de los ficheros de informaci3n p3blica 1.35*, https://www.omie.es/sites/default/files/2024-06/Formato_ficheros_inf_pub_135_0.pdf, Accessed on May 27, 2025, Jun. 2024.